

Vol 10 Chapter 6 — Group Theory and Lie Groups

Keeper (author pass — deep math/physics revision)

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Chapter 6 — Group Theory and Lie Groups

Level 1 — one sentence

Groups capture symmetry, Lie groups are smooth-manifold groups with smooth operations, and the BST framework is built on the specific Lie group $SO_0(5, 2)$ — its 21-dimensional real Lie algebra $\mathfrak{so}(5, 2)$, maximal compact subgroup $SO(5) \times SO(2)$, rank-2 Cartan subalgebra with half-sum of positive roots $\rho = (5/2, 3/2)$, and bounded symmetric domain coset $D_{IV}^5 = SO_0(5, 2)/[SO(5) \times SO(2)]$.

Level 2 — graduate-physicist precision

6.1 Groups and subgroups

A group $(G, \cdot)$: set with associative operation, identity, inverses. Examples: $\mathbb{Z}$ (additive), S_n (permutations), $O(n)$ (orthogonal matrices), $SU(2)$ (unitary 2×2 , $\det = 1$).

Subgroups: $H \subset G$ closed under $\cdot$. Quotient G/H for normal subgroups.

6.2 Lie groups

A Lie group is a smooth manifold G with smooth group operations (multiplication and inversion). Examples: $O(n)$, $SO(n)$, $SU(n)$, $Sp(n)$, $GL(n, \mathbb{R})$, $\mathbb{R}^n$ (translation), S^1 (rotation).

Lie algebra $\mathfrak{g} = T_e G$ = tangent space at identity. Comes with Lie bracket $[X, Y] = XY - YX$ (for matrix groups) satisfying antisymmetry + Jacobi identity.

Exponential map: $\exp : \mathfrak{g} \rightarrow G$, e.g., $e^A = \sum \frac{A^k}{k!}$ for matrix Lie algebra.

6.3 Cartan classification

Cartan classified simple Lie algebras (1894): - $A_n = \mathfrak{su}(n+1)$ (special unitary) - $B_n = \mathfrak{so}(2n+1)$ (odd orthogonal) - $C_n = \mathfrak{sp}(2n)$ (symplectic) - $D_n = \mathfrak{so}(2n)$ (even orthogonal) - G_2, F_4, E_6, E_7, E_8 (exceptional)

Plus reductive (semisimple + abelian).

6.4 $SO(5, 2)$ and D_{IV}^5

For BST: $SO(5, 2)$ = orthogonal group of indefinite signature $(5, 2)$. $SO_0(5, 2)$ = identity component (connected).

$\mathfrak{so}(5, 2)$: rank 2; $\dim = 21$ over $\mathbb{R}$. Maximal compact subgroup $SO(5) \times SO(2)$ has $\dim 10 + 1 = 11$.

Quotient $D_{IV}^5 = SO_0(5, 2)/[SO(5) \times SO(2)]$ is the Type IV bounded symmetric domain of complex dimension 5 (real dimension 10).

Half-sum of positive roots: $\rho = (5/2, 3/2)$. These specific Cartan parameters propagate through BST mass formulas, the Bergman normalization $c_{FK} \pi^{9/2} = 225$, and the operator zoo (Vol 5 Ch 1).

6.5 Representations and characters

Group representation: homomorphism $\rho : G \rightarrow GL(V)$. Irreducible reps are building blocks; semisimple groups have completely reducible reps.

Character $\chi(g) = \text{Tr } \rho(g)$: invariant under conjugation, complete invariant for rep.

For compact Lie groups: characters are determined by their values on maximal torus (Weyl character formula).

6.6 K-audit anchors

- Vol 5 Ch 1: BST substrate group structure
- Vol 5 Ch 3: $SO(5)$ angular momentum
- Vol 11 Ch 4-5: deeper treatment of D_{IV}^5 geometry

Level 3 — 5th-grader accessibility

Groups describe symmetries. Lie groups are smooth-manifold groups (rotations, Lorentz transformations, gauge groups). Lie algebras are the infinitesimal versions — easier to compute with. Cartan classified them all in 1894 into four infinite families + 5 exceptional groups. BST's group is $SO_0(5, 2)$: 21-dimensional, contains Lorentz + extra dimensions. Its quotient by the maximal compact subgroup gives D_{IV}^5 — the bounded symmetric domain where all of BST's physics lives.

What comes next

Chapter 7 develops differential geometry.

Where to look this up

- Fulton and Harris, *Representation Theory*
- Knapp, *Lie Groups Beyond an Introduction*
- BST: Vol 5 Ch 1; Vol 11 Ch 4-5